

also be selected using hexadecimal colour codes as used in HTML colour selection. The GIMP has native support for indexed colour and RGB colour spaces; other colour spaces are supported using decomposition, with which each channel of the new colour space becomes a black-and-white image. CMYK, LAB and HSV (hue, saturation and value) are supported this way. Colour blending can be achieved using the blend tool, by applying a gradient to the surface of an image and using the GIMP's colour modes. Gradients are also integrated into tools such as the brush tool, so when the user paints this way, the output colour slowly changes. There are a number of default gradients included with the GIMP; a user can also create custom gradients with the tools provided. Gradient plugins are also available.

Selections and paths

The GIMP offers a rectangular and circular selection tool, a free select tool, and fuzzy select tool (also known as magic wand). More advanced options include the 'select by colour' tool for selecting contiguous regions of colour, and the 'scissors select' tool, which creates selections semi-automatically between areas of highly contrasting colours. The GIMP also supports a quick mask mode with which you can use a brush to paint the area of a selection. This looks like a red coloured overlay being added or removed. The foreground select tool is an implementation of Simple Interactive Object Extraction (SIOX), a method to extract foreground elements, such as a person or a tree that is in focus. The 'paths' tool allows you to create vectors (also known as Bézier curves). You can use paths to create complex selections, including around natural curves. You can then paint (or 'stroke') the paths with brushes, patterns, or various line styles; and name and save paths for reuse.

Image editing

There are many tools that can be used for editing images in the GIMP. The more common tools include a paint brush, pencil, air brush, eraser and ink tools to

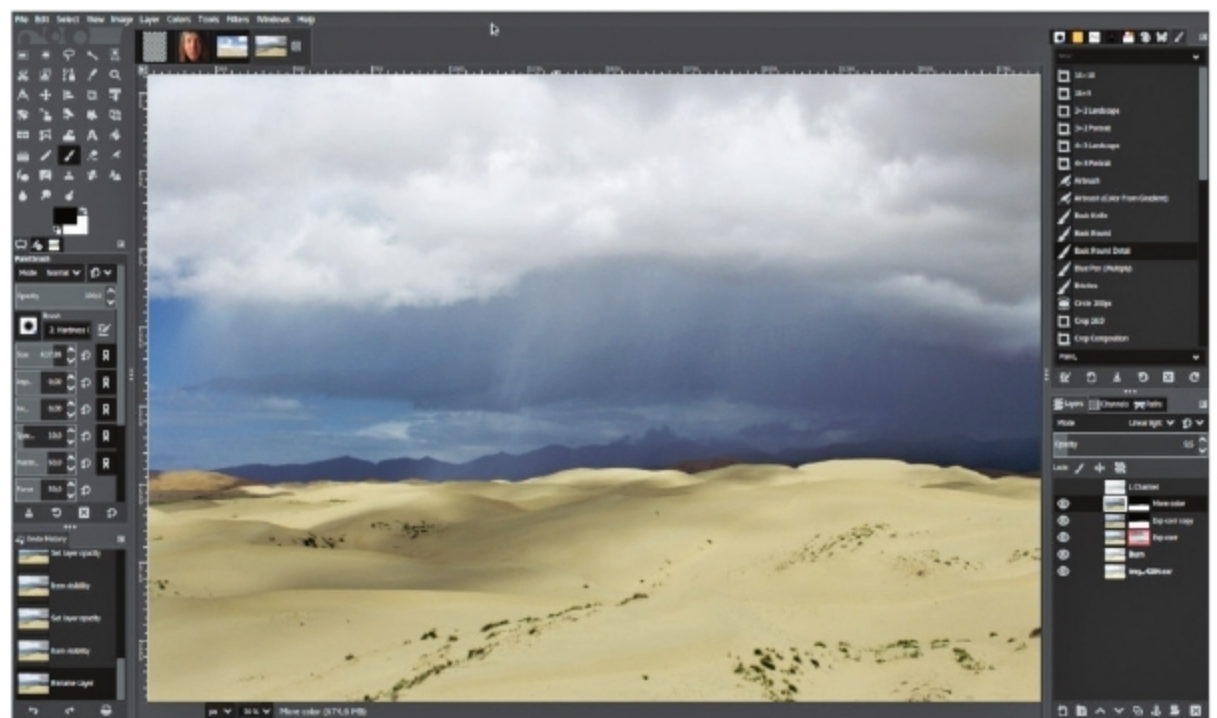


Figure 1: GIMP 2.10 new dark theme



Figure 2: GIMP icons in four different themes

create new or blended pixels. The bucket filling tool can be used to fill a selected area with a colour or pattern. The blend tool can be used to fill a selected area with a colour gradient. These colour

transitions can be applied to large regions or smaller custom path selections.

The GIMP also provides 'smart' tools that use a more complex algorithm to do things that otherwise would be